



Universiteit Antwerpen
| Faculteit Wetenschappen

Compilers project

2025-2026

Practical

- Room: M.G.026 (online during March!)
- Time: 14:00-16:00 (March: TBD, see Blackboard)
- Cancelled lab sessions
- Goal: Interactive Q&A
- Overview PDF
- Extra questions:
 - Theory + exam: Guillermo (guillermo.perez@uantwerpen.be)
 - Project: Kasper (kasper.engelen@uantwerpen.be)
 - Grading: Guillermo and Kasper

Project overview

- **Learn to write your own compiler**
 - Groups of 3
- **Targets**
 - LLVM IR
 - MIPS (existing tooling)
 - Native binary (x86, ARM) (existing tooling)
- **Features: mandatory and optionals**
- **Optimisations**
- **Visualisations**

Assignments

1. **Basic parsing: expressions, whitespace, ...**
2. **Types and variables**
3. **Comments, Arrays, strings, printf, scanf, LLVM IR**
4. **Loops, if-statements, enums**
5. **Functions, headers**
6. **Advanced features: structs, files, dynamic memory**
7. **Compilation to assembly: x86, ARM, MIPS**

Also: custom features (after approval!)

Technical information

- **ANTLR: overview PDF**
- **LLVM lite**
- **LLVM tools**
- **Tests: zip files**

Recommended schedule

- Big project
- Don't waste time
- Deadlines:
 - These are indications, the exact date will be announced via Blackboard!
 - Feb 20 (next week): sign up for a group
 - March 20: projects 1-3
 - May 8: full LLVM IR
 - May 29: binary and MIPS output
- Recommended schedule (see table)

Date	Topic
13 Feb	Intro + 1 expr
20 Feb	1 Expr + 2 variables
27 Feb	2 Variables
6 March	3 Arrays, I/O
13 March	3 Arrays, I/O
20 March	4 Loops
27 March	5 Functions
3 April	5 Functions
24 April	6 Advanced
8 May	6 Advanced
16 May	7 LLVM -> Bin
23 May	7 LLVM -> Bin

Submitting and grading

- **Grade awarded to group**
 - No group = no grading.
 - Deadline for Blackboard groups: February 20
- **Demo video**
- **Blackboard: code + link to video**
- **Grading:**
 - 90%: LLVM IR
 - Mandatory = required (counts for half of the 90%)
 - Optional = extra (counts for half of the 90%)
 - 10%: Assembly/Binary
 - All or nothing
 - Just mandatory LLVM IR is not enough!

